

DSA Logic Notebook (Pattern Wise)

Sliding Window

- Used when: Subarray / Substring (contiguous)
- Key idea: Expand right, shrink left
- Time Complexity: $O(n)$
- Common mistakes: Forgetting to shrink, wrong condition order

Two Pointers

- Used when: Sorted arrays, comparisons
- Key idea: Move pointers strategically
- Time Complexity: $O(n)$
- Common mistakes: Infinite loops, moving wrong pointer

Hashing

- Used when: Frequency, counting, lookup
- Key idea: Trade space for speed
- Time Complexity: $O(n)$
- Common mistakes: Forgetting updates, unnecessary HashMaps

Stack

- Used when: Previous/Next greater or smaller, validation
- Key idea: LIFO, monotonic behavior
- Time Complexity: $O(n)$
- Common mistakes: Wrong pop condition, empty stack handling

Binary Search

- Used when: Sorted data or monotonic answer
- Key idea: Divide search space
- Time Complexity: $O(\log n)$
- Common mistakes: Off-by-one, infinite loops

Recursion

- Used when: Self-similar problems

- Key idea: Trust recursive call
- Common mistakes: Missing base case, stack overflow

Backtracking

- Used when: All combinations/permutations
- Key idea: Choose, explore, un-choose
- Common mistakes: Forgetting to backtrack

Greedy

- Used when: Optimization problems
- Key idea: Best choice at each step
- Common mistakes: No proof of correctness

Dynamic Programming

- Used when: Overlapping subproblems
- Key idea: Store and reuse
- Steps: State, Transition, Base case
- Common mistakes: Wrong state definition

Graph (BFS/DFS)

- Used when: Connections, paths
- BFS: Shortest path, DFS: Depth exploration
- Common mistakes: Missing visited array

Heap / Priority Queue

- Used when: Top K, min/max problems
- Key idea: Best element on top
- Common mistakes: Wrong heap type

Bit Manipulation

- Used when: XOR, subsets, parity
- Key idea: Bits instead of extra space
- Common mistakes: Wrong bit operations